Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-2949
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. The sections below document the testing approach, tools used, and results obtained for the project Auto Ticket Classification using Flow Designer.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6 (with Listeners: Summary Report, Aggregate Report, Response Time Graph)
Type of Testing	Load Testing, Stress Testing and Spike Testing
Target Module / API	Incident creation via ServiceNow REST Table API (POST /api/now/table/incident), which triggers the Flow Designer flow "Auto Ticket Classification" (category / assignment group / priority assignment)
Test Environment	Cloud hosted – ServiceNow Personal Developer Instance (PDI)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load – users create incidents with typical short descriptions (e.g., "VPN not connecting"); flow classifies each ticket	10	60	All tickets classified correctly; avg response < 2 sec; 0% errors
2	Normal Load – steady stream of incident creation across Hardware, Software, Network and Access categories	25	120	Flow executes for every ticket; error rate < 1%
3	Peak Load – high concurrent incident creation simulating a busy support period	50	180	Avg response < 2 sec; max < 5 sec; CPU and memory < 80%
4	Spike Test – sudden surge of incident creation within a short ramp-up time	100	60	System stays available; error rate < 1%; flow queue recovers after spike

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Within target across all scenarios
2	Response Time (Max)	< 5 seconds	3.87 seconds	Pass	Highest value seen during spike test
3	Throughput (Req/sec)	≥ 10 req/sec	14.6 req/sec	Pass	Stable during peak load
4	Error Rate	< 1%	0.4%	Pass	Few timeouts at 100 users
5	CPU Utilization	< 80%	62%	Pass	Peak value during spike test
6	Memory Utilization	< 80%	68%	Pass	No memory growth after test

Step 4: Observations & Analysis

Key Findings:

The Auto Ticket Classification flow classified incidents correctly and consistently under all four test scenarios. Average response time stayed at 1.42 seconds and the maximum at 3.87 seconds, both within target. Throughput was stable at about 14.6 requests per second with an overall error rate of 0.4%. CPU (62%) and memory (68%) stayed below the 80% limit.

Bottlenecks Identified:

1. Response time rose slightly and a few requests timed out during the 100-user spike because many flow executions queued at the same time.
2. The flow ran several lookup steps (category and assignment group) one after another, which added to processing time under peak load.
3. The PDI has limited resources, so heavy concurrent use affects performance.

Optimization Steps Taken:

1. Reduced redundant Look Up Record actions in the flow and used a single decision table for category and assignment group mapping.
2. Added indexes on the fields used in the flow conditions.
3. Set the flow to run asynchronously to avoid blocking incident creation.
4. Increased the JMeter ramp-up time and added a timer between requests to reduce unnecessary spikes. The test was re-run and the error rate improved.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]